

Xiao Fu

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EDUCATION

Zhejiang University

Hangzhou, China

Bachelor in Information Engineering | College of Information Science & Electronic Engineering Sept. 2018 – Jul. 2022

- **GPA:** 3.96/4.00, 90.53/100, **Ranking:** 4/145
- **National Scholarship** awardee, **Twice**
- **Courses:** Computer Vision(96), Digital Signal Processing(95), Matrix Theory(94), Stochastic Process(96), Probability and Mathematical Statistics(95), Numerical Analysis(92), Signals and Systems(98), Calculus(A)(90)

PUBLICATIONS

* indicates equal contribution

- [2] [Xiao Fu*](#), Shangzhan Zhang*, Tianrun Chen, Yichong Lu, Lanyun Zhu, Xiaowei Zhou, Andreas Geiger, Yiyi Liao, “**Panoptic NeRF: 3D-to-2D Label Transfer for Panoptic Urban Scene Segmentation**,” *Under Review* [[Paper](#) | [Project](#) | [Code](#)]
- [1] Guangyi Zhang*, [Xiao Fu*](#), Qiyu Hu, Yunlong Cai, Guanding Yu, “**Hybrid Precoding Design Based on Dual-Layer Deep-Unfolding Neural Network**,” *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Helsinki, Finland, 2021, pp. 678-683 [[Paper](#) | [Code](#) | [Slides](#)]

RESEARCH EXPERIENCE

Panoptic NeRF: 3D-to-2D Panoptic Label Transfer

Max Planck Institute for Intelligent Systems

Supervisor: [Prof. Andreas Geiger](#) and [Prof. Yiyi Liao](#)

Sept. 2021 – Mar. 2022

- Propose Panoptic NeRF, which aims for obtaining per-pixel 2D semantic and instance labels from easy-to-obtain coarse 3D bounding primitives. Our method utilizes NeRF as a differentiable tool to unify coarse 3D annotations and 2D semantic cues transferred from existing datasets. This combination allows for improved geometry guided by semantic information which enables rendering of accurate semantic maps across multiple view. By inferring in 3D space and rendering to 2D labels, our 2D semantic and instance labels are multi-view consistent by design

Algorithm-Based Dual-Layer Deep-Unfolding Neural Network

Zhejiang University

Supervisor: [Prof. Guanding Yu](#)

Jun. 2020 – Feb. 2021

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network (DLDUNN) which unfolds the iterative Penalty Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award (Team leader) - Student Research Training Program (Province-level)**

HONORS AND SCHOLARSHIPS [[Certificate & Report](#)]

- National Scholarship (**Highest Honor for Chinese Undergraduates, Twice**) 2020, 2021
- ISEE Honor Student Award (**Top 5 students among ISEE 314 undergraduates**) 2021
- ISEE Yuelun Alumni Scholarship (**Top 10 teams out of ISEE undergraduates and graduates**) 2020, 2021
- First-Class Scholarship for Academic Excellence of Zhejiang University (**Top 3%**) 2020, 2021
- Government Scholarship for Academic Excellence of Zhejiang Province (**Top 3%**) 2019
- Outstanding Graduation Thesis of Zhejiang University 2022
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 7% out of 26,112 teams**) 2021
- Gold Prize – 7th International “Internet+” Innovation and Entrepreneurship Contest (**Team Leader**) 2021
- Gold Prize – 12th “Challenge Cup” National College Student Business Plan Competition 2020
- Grand Prize – “Yongdian Cup” Innovation and Entrepreneurship Contest (**Team Leader**) 2019
- Grand Prize – “Midea” Global Technology Innovation Contest 2020
- Academic Excellence Scholarship of Zhejiang University 2019, 2020, 2021
- Zhejiang University Five-Star Volunteer 2021